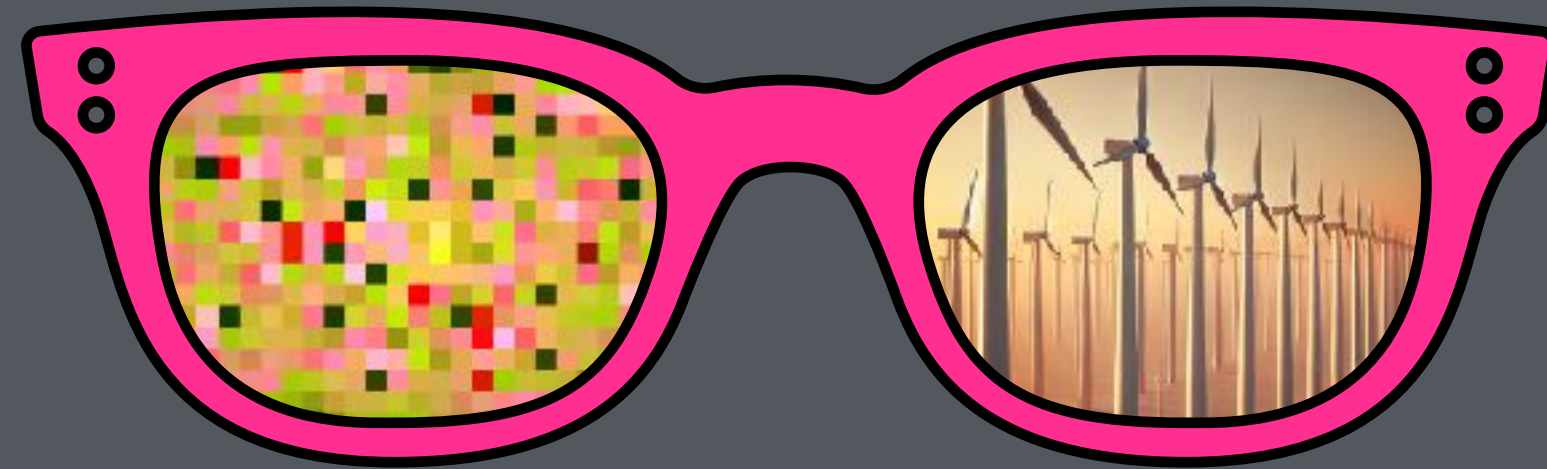


Data and the State

DATA 20195 / 30195



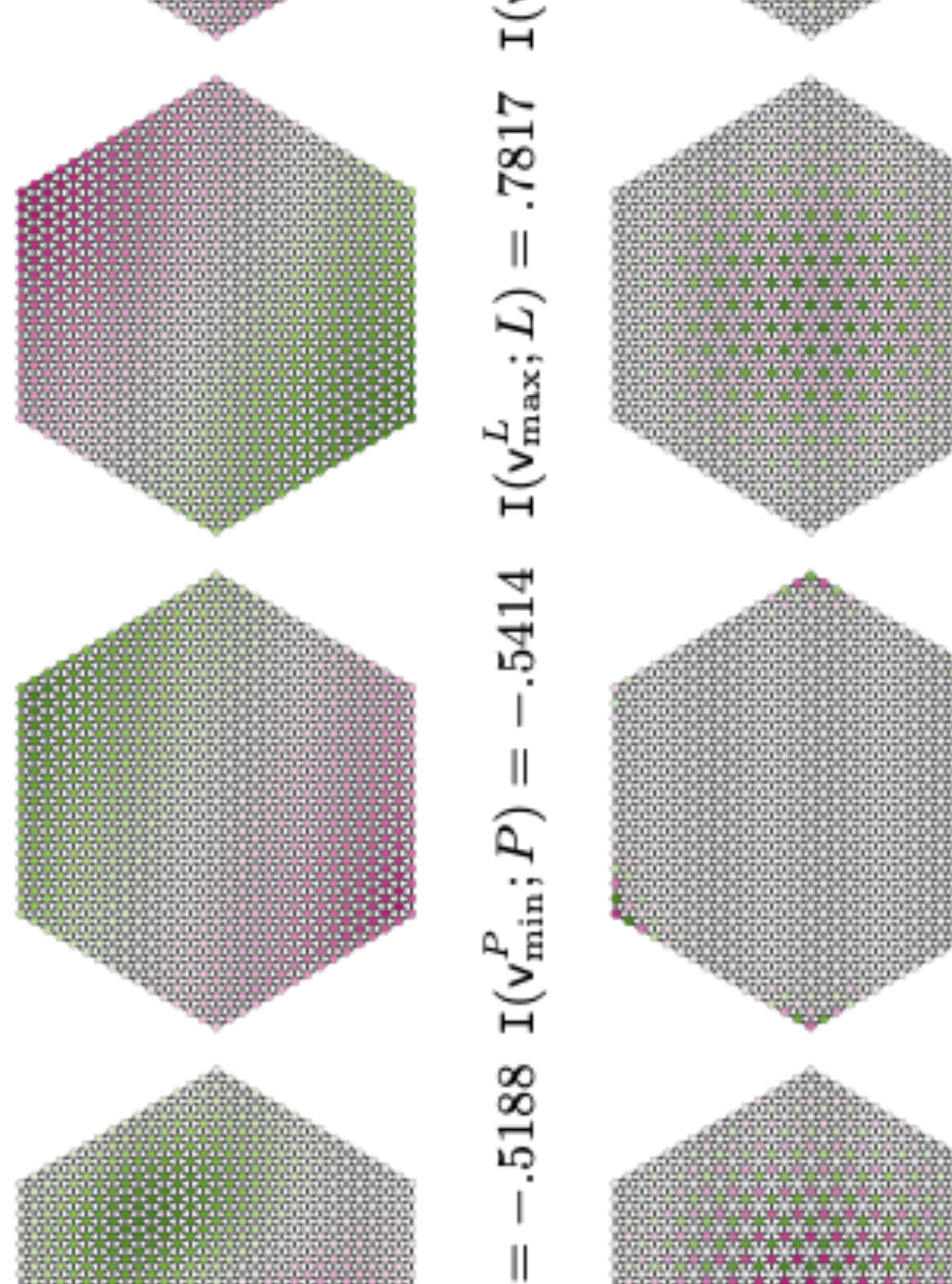
Space and measurement

Lecture 14, Wednesday May 6

What is segregation?

- We want to understand how to think about, and how to measure, **segregation**
- Broadly, segregation is **separation** — can refer to public services, amenities, etc
- Possible working definition of housing segregation: when subpopulations live separately within a common geography
- We'll look at some quantitative definitions

$$= 1.0265 \quad I(v_{\max}^P; P) = .9986 \quad I(v_{\min}^L; L) = .0022 \quad I(v_{\max}^L; L) = .7817 \quad I(v_{\min}^P; P) = -.5414 \quad I(v_{\min}^L; L) = -.5188$$



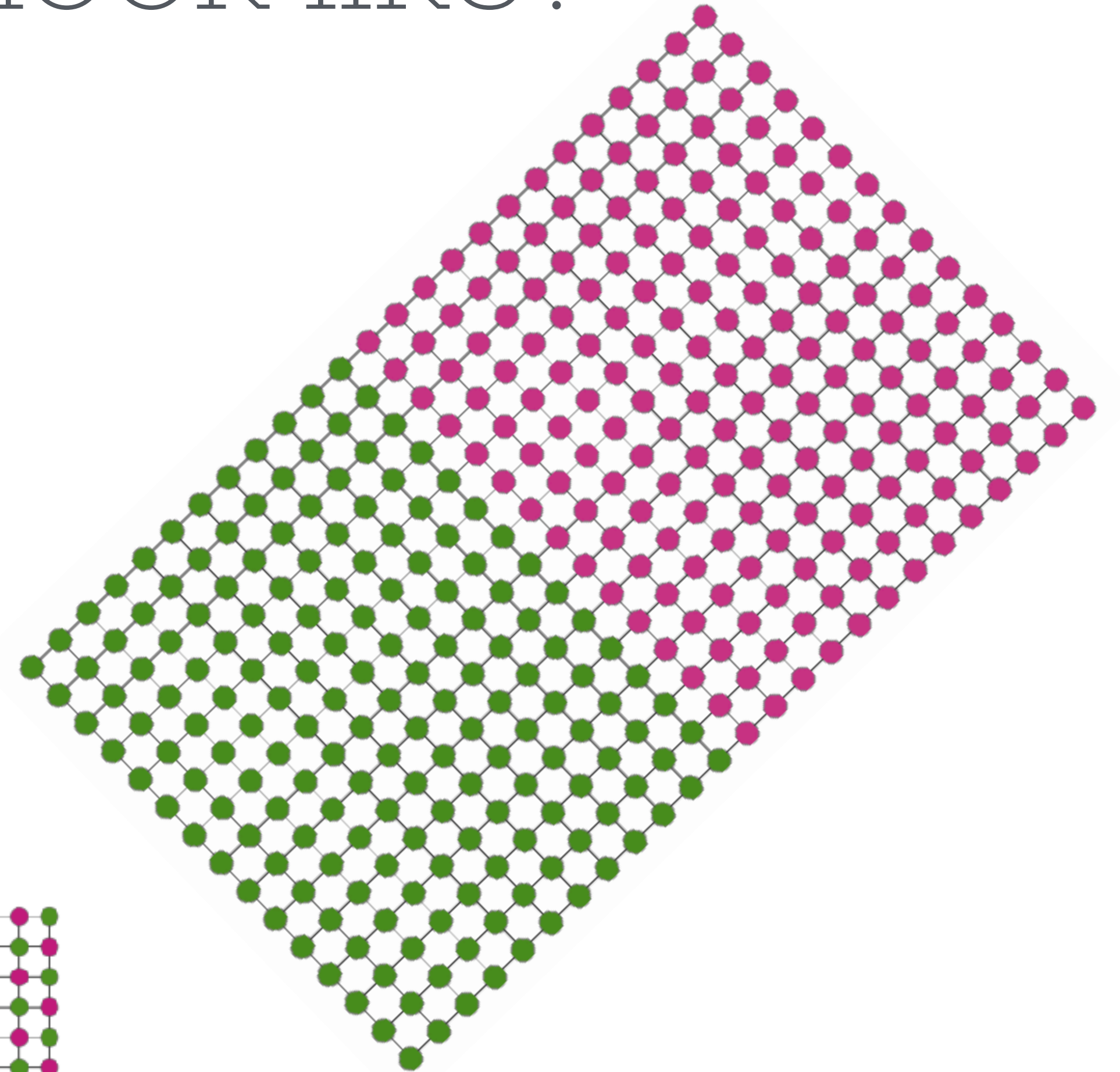
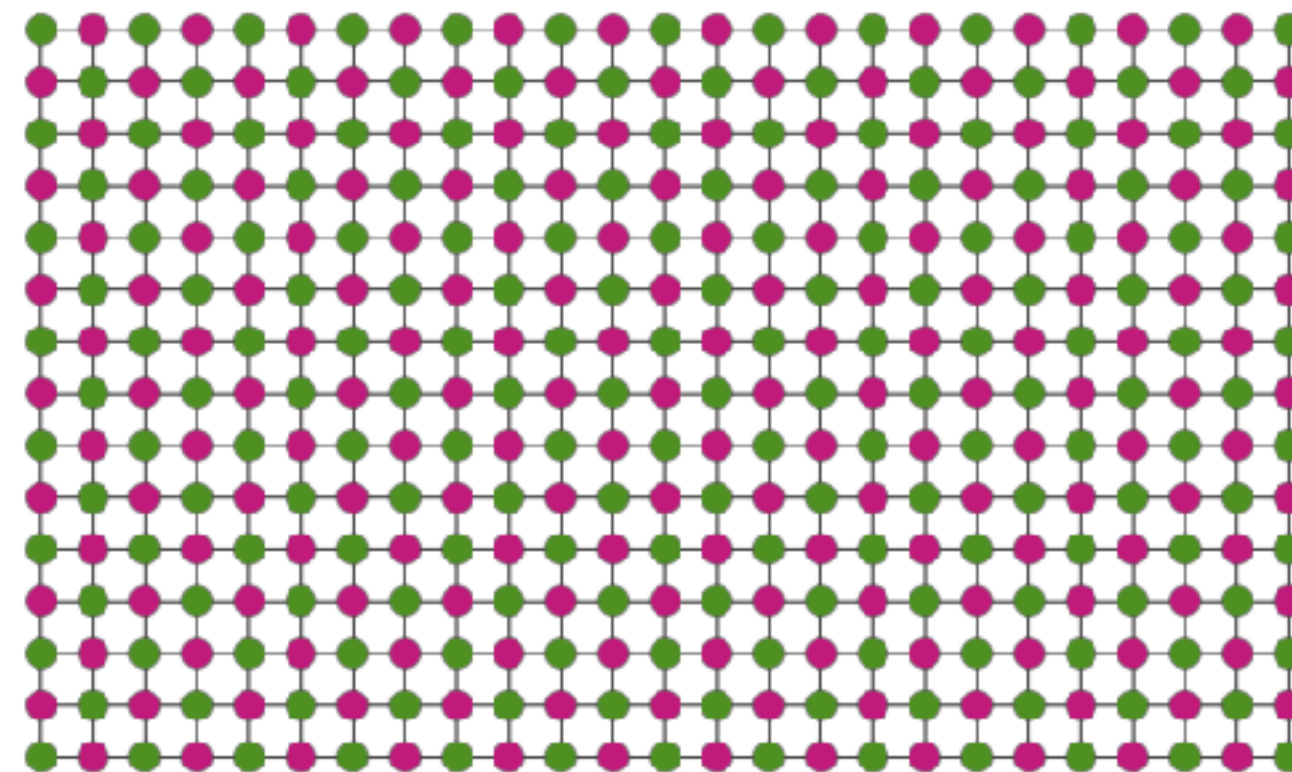
What does segregation look like?

Suppose you have a group **X** and its complement **Y**

Do they live interspersed? Clustered?

Units matter, scale matters!

recall MAUP - modifiable areal unit problem



Tuskegee example

1954: Brown v. Board of Ed mandates integration, speeds civil right movement

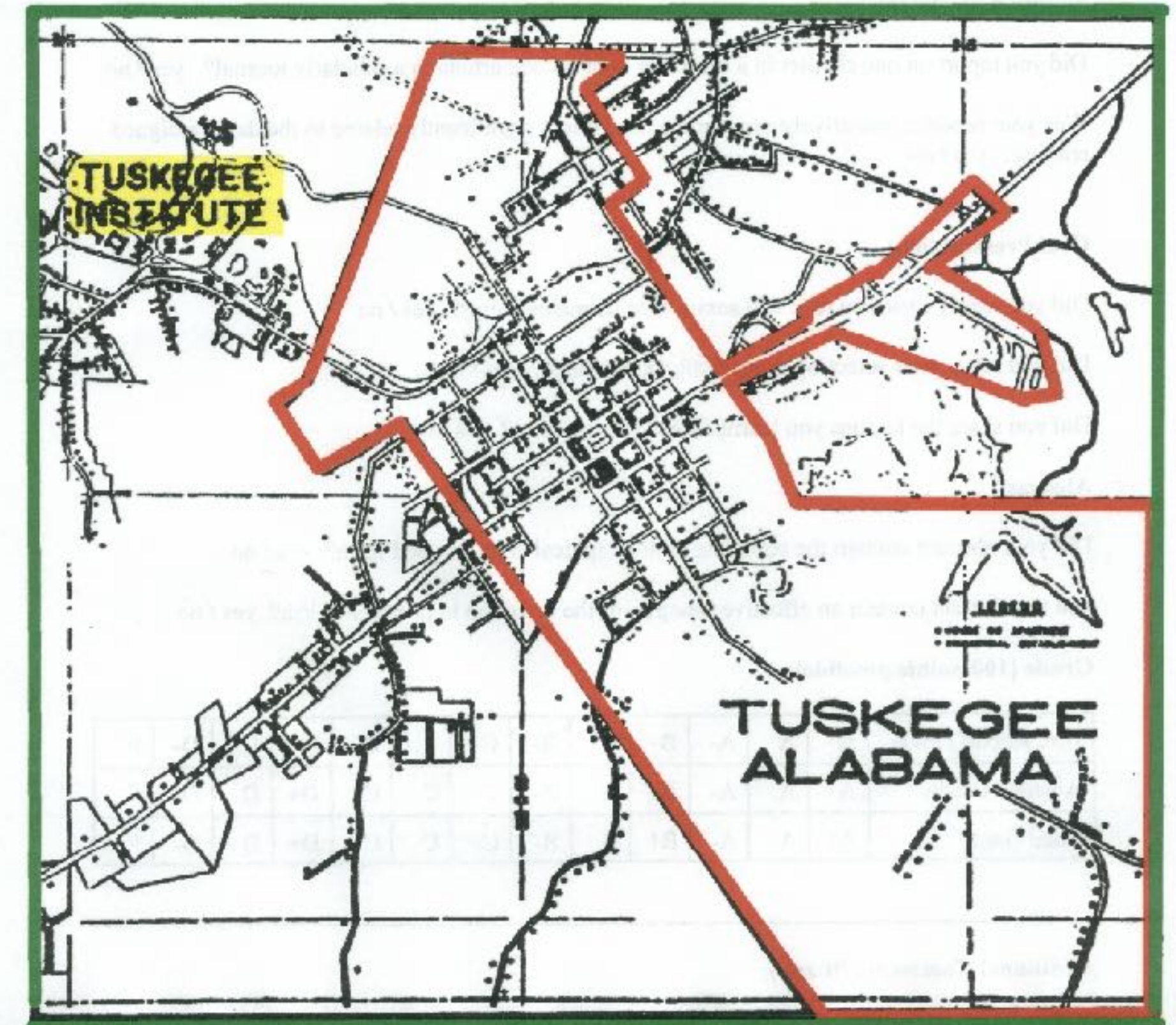
1957: Tuskegee, Alabama redrawn by Leg.

Old (**square**) — 80% Black

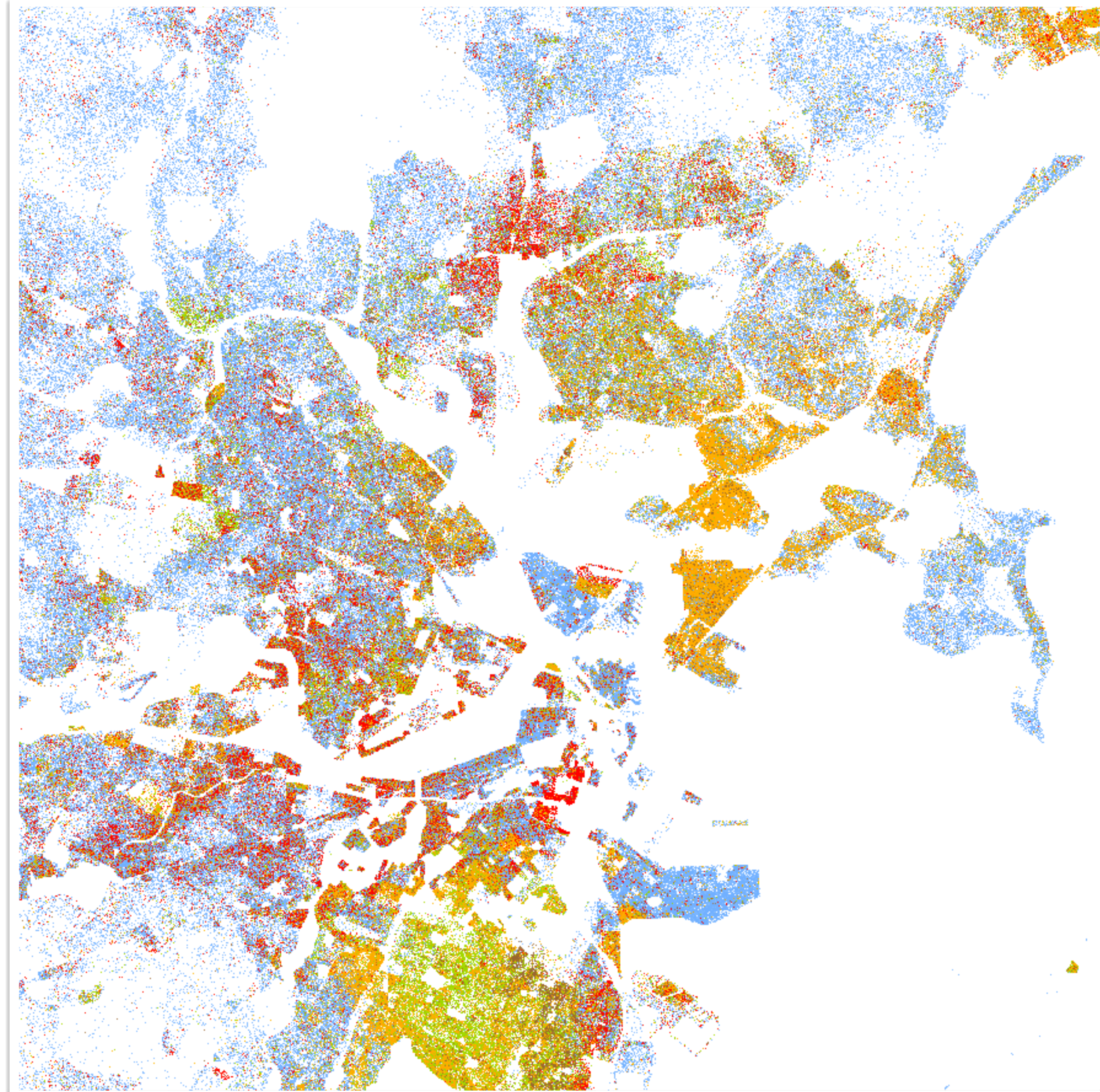
New (**28-sided polygon**) — 100% White

Supreme Court intervenes in Gomillion v. Lightfoot (**1960**)

APPENDIX TO OPINION OF THE COURT.
CHART SHOWING TUSKEGEE, ALABAMA, BEFORE AND AFTER ACT 140



(The entire area of the square comprised the City prior to Act 140. The irregular black-bordered figure within the square represents the post-enactment city.)

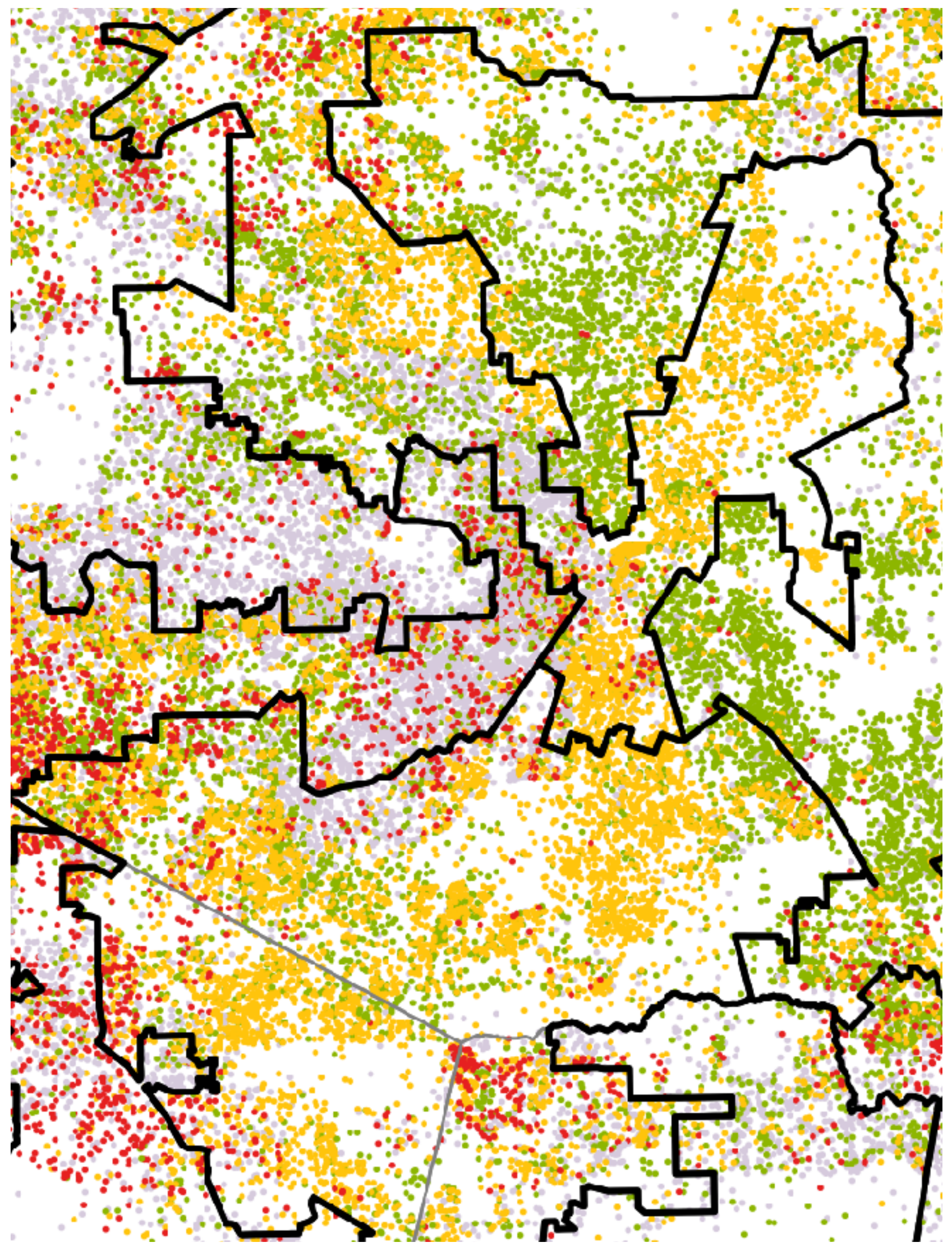


2010 Census Block Data

1 Dot = 1 Person

- White
- Black
- Asian
- Hispanic
- Other Race / Native American / Multi-racial

Boston area

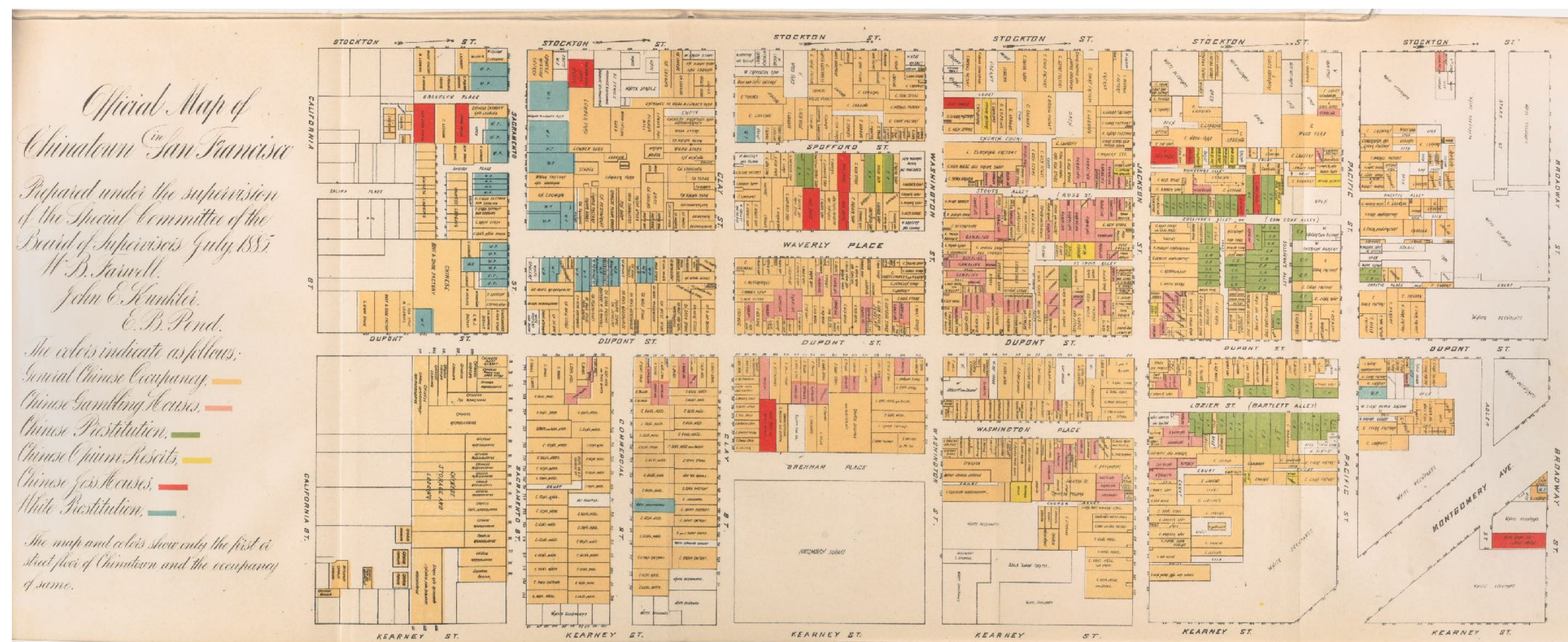


Houston area

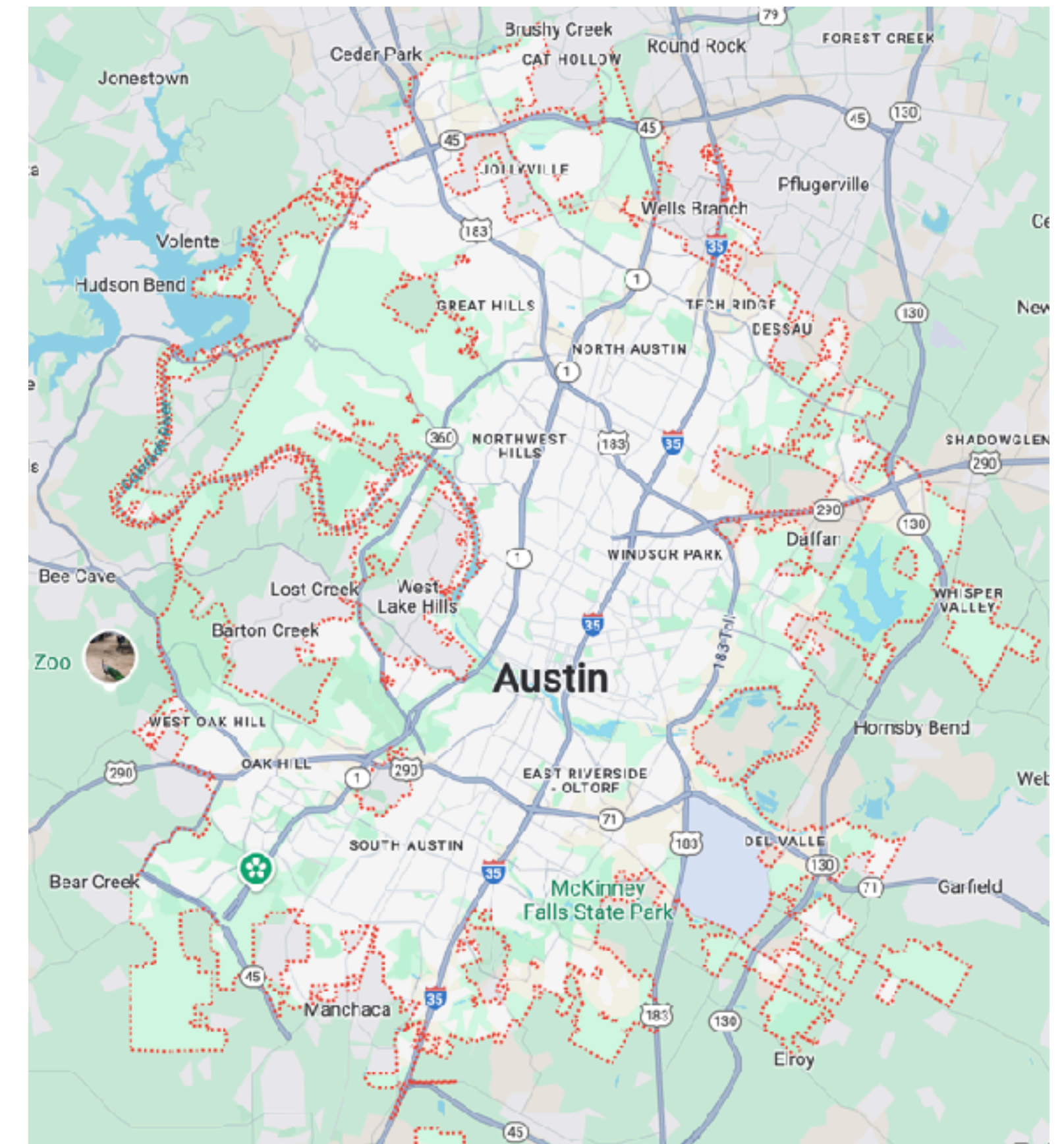
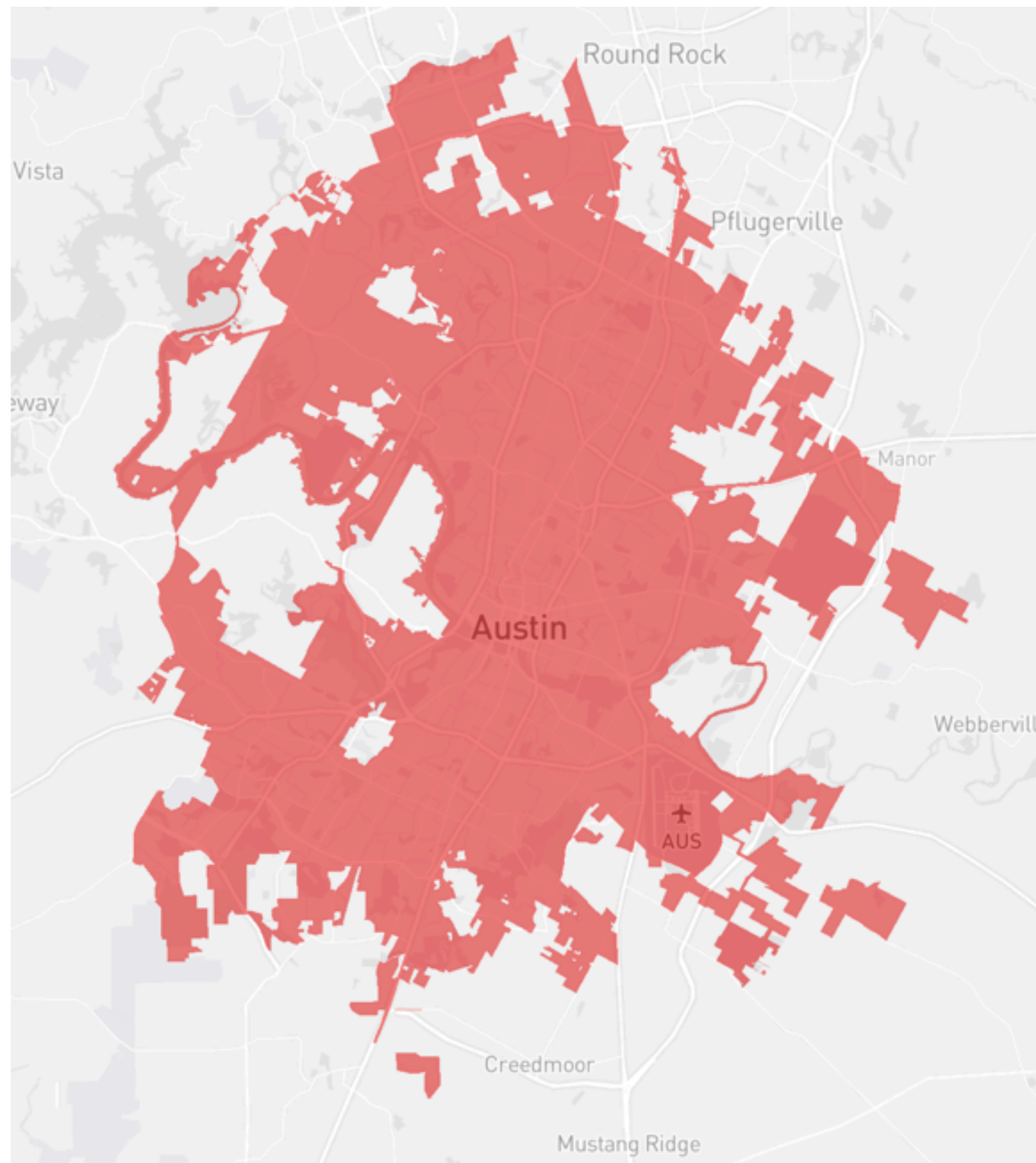
mechanisms vary

Example: Asian segregation

- Alien Land Laws: Beginning with California (1913) and eventually adopted by many western states, these laws prohibited "aliens ineligible for citizenship" (which included all people of Chinese ancestry until 1943) from owning land or property.
- San Francisco Cubic Air Ordinance (1870): Required 500 cubic feet of air per person in living quarters, **selectively enforced** against Chinese residences to criminalize their housing arrangements. — coupled with Laundry Ordinance, Queue Ordinance, etc
- Residential pressure and restrictions
 - Restrictive covenants in property deeds prohibiting sale to "Mongolians"
 - Neighborhood improvement associations with explicit anti-Chinese policies
 - Municipal zoning that concentrated laundries and other Chinese businesses
 - Police non-protection



Example: Austin, TX



I-35 (late 50s/early 60s)

metrics matter

Metrics of segregation

Suppose you have a group X and its complement Y

Have counts or shares x_i at every node of a network

Let's let $\bar{x}$ be the *average* of the shares. Let $\mathbf{v}$ be the zero-centered vector ($v_i = x_i - \bar{x}$).

- **Dissimilarity**: average difference $x_i - \bar{x}$
- **Gini**: average difference $x_i - x_j$

- **Moran's I**:
$$I(\mathbf{x}) = \frac{\mathbf{v}^T W \mathbf{v}}{\mathbf{v}^T \mathbf{v}}$$
  add up products of neighboring values

This view is verified by historical studies that report quantitative indices of racial segregation. The standard measure of segregation is the index of **dissimilarity**, which captures the degree to which blacks and whites are evenly spread among neighborhoods in a city.⁶ Evenness is defined with respect to the racial composition of the city as a whole. If a city is 10% black, then an even residential pattern requires that every neighborhood be 10% black and 90% white. Thus, if a neighborhood is 20% black, the excess 10% of blacks must move to a neighborhood where the black percentage is under 10% to shift the residential configuration toward evenness. The index of dissimilarity gives the percentage of blacks who would have to move to achieve an “even” residential pattern—one where every neighborhood replicates the racial composition of the city.

- **Assortativity** (“copy”): share of XX edges
- **Assortativity'**: share of XX half-edges

Dissimilarity

- **Dissimilarity**: differences x_i vs $\bar{x}$

This view is verified by historical studies that report quantitative indices of racial segregation. The standard measure of segregation is the index of **dissimilarity**, which captures the degree to which blacks and whites are evenly spread among neighborhoods in a city.⁶ Evenness is defined with respect to the racial composition of the city as a whole. If a city is 10% black, then an even residential pattern requires that every neighborhood be 10% black and 90% white. Thus, if a neighborhood is 20% black, the excess 10% of blacks must move to a neighborhood where the black percentage is under 10% to shift the residential configuration toward evenness. The index of dissimilarity gives the percentage of blacks who would have to move to achieve an “even” residential pattern—one where every neighborhood replicates the racial composition of the city.

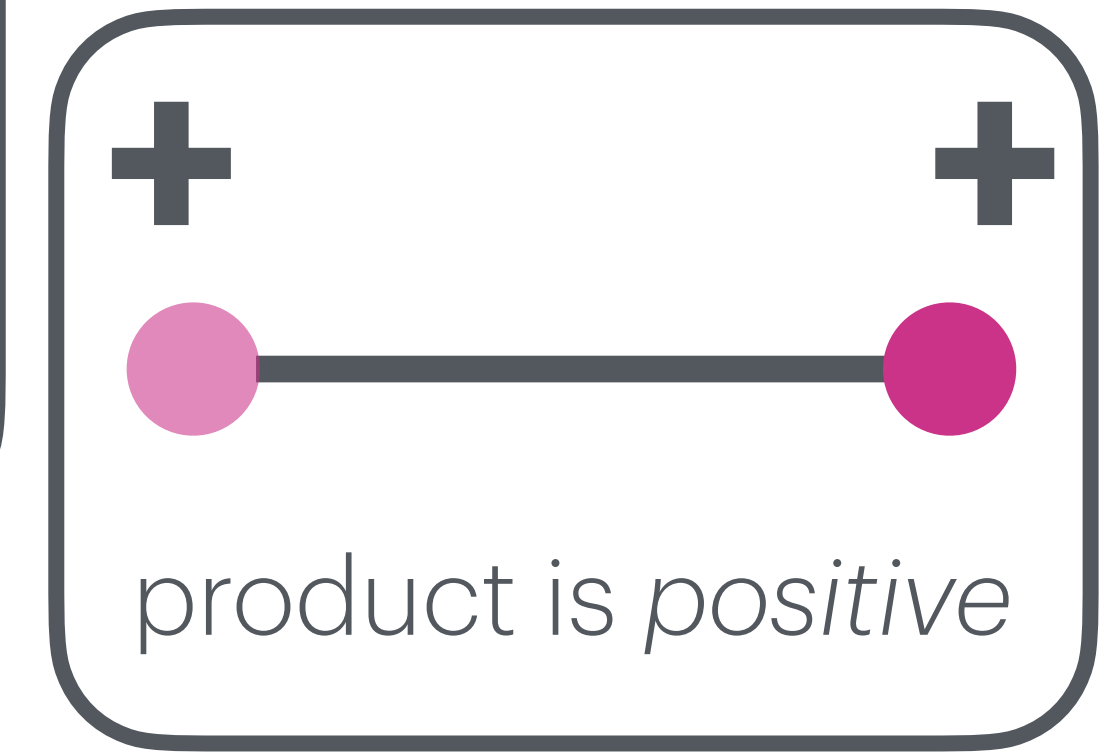
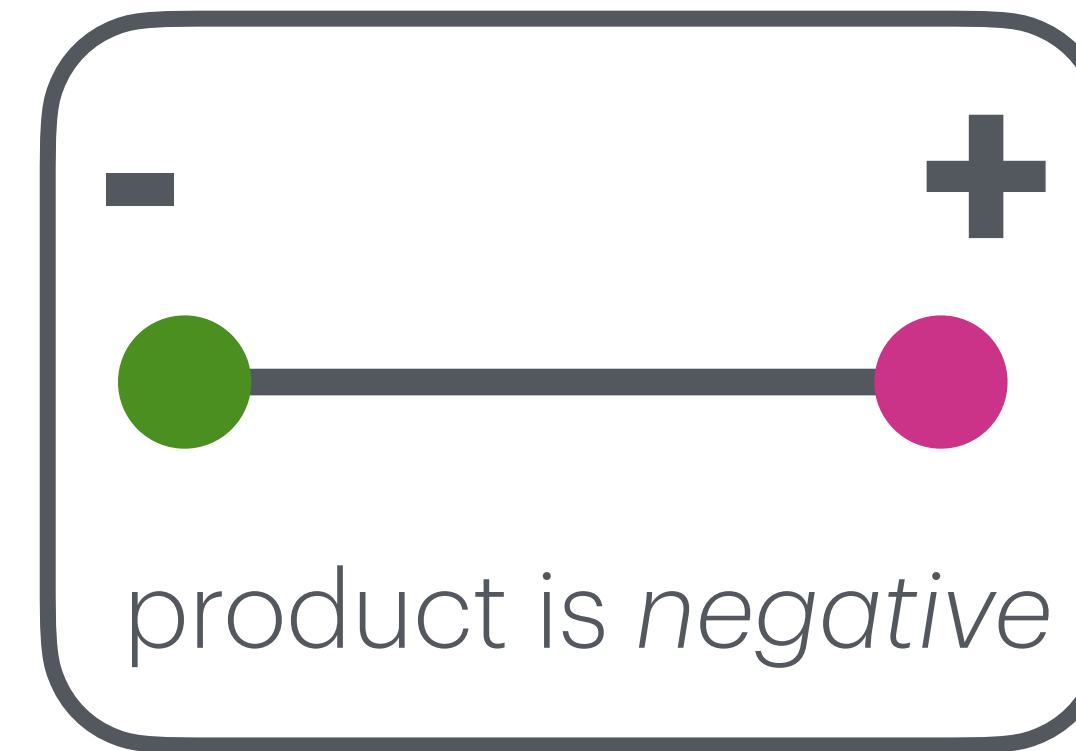
from Massey, *American Apartheid*

Table 2.1 Indices of black-white segregation (dissimilarity) in selected northern and southern cities: circa 1860–1870, 1910, and 1940

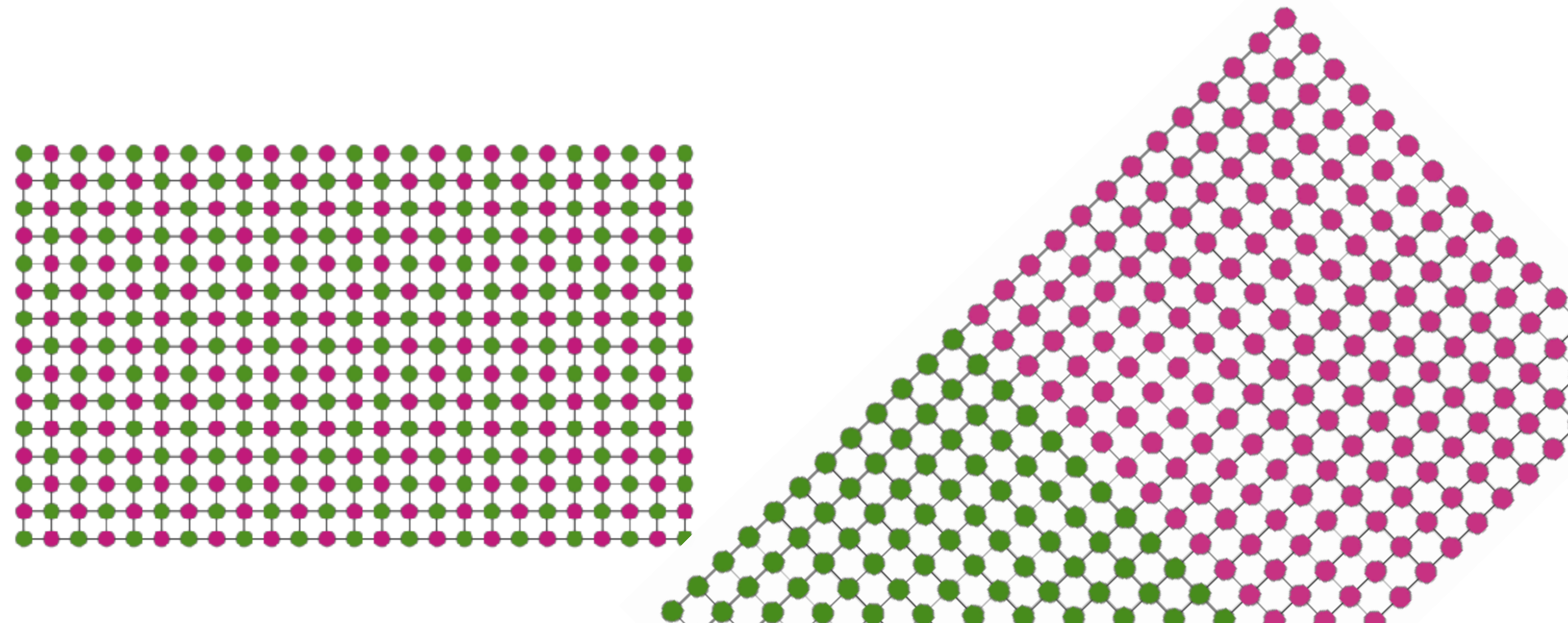
	Free blacks vs. whites, circa 1860	Blacks vs. native whites, 1910	Nonwhites vs. whites, 1940
<i>Northern cities</i>			
Boston	61.3	64.1	86.3
Chicago	50.0	66.8	95.0
Cincinnati	47.9	47.3	90.6
Cleveland	49.0	69.0	92.0
Indianapolis	57.2	—	90.4
Milwaukee	59.6	66.7	92.9
New York	40.6	—	86.8
Philadelphia	47.1	46.0	88.8
St. Louis	39.1	54.3	92.6
San Francisco	34.6	—	82.9
Wilmington	26.1	—	83.0
Average	45.7	59.2	89.2
<i>Southern cities</i>			
Augusta	—	58.8	86.9
Baltimore	22.1	—	90.1
Charleston	23.2	16.8	60.1
Jacksonville	—	39.4	94.3
Louisville	20.2	—	81.7
Mobile	29.8	—	86.6
Nashville	43.1	—	86.5
New Orleans	35.7	—	81.0
Average	29.0	38.3	81.0

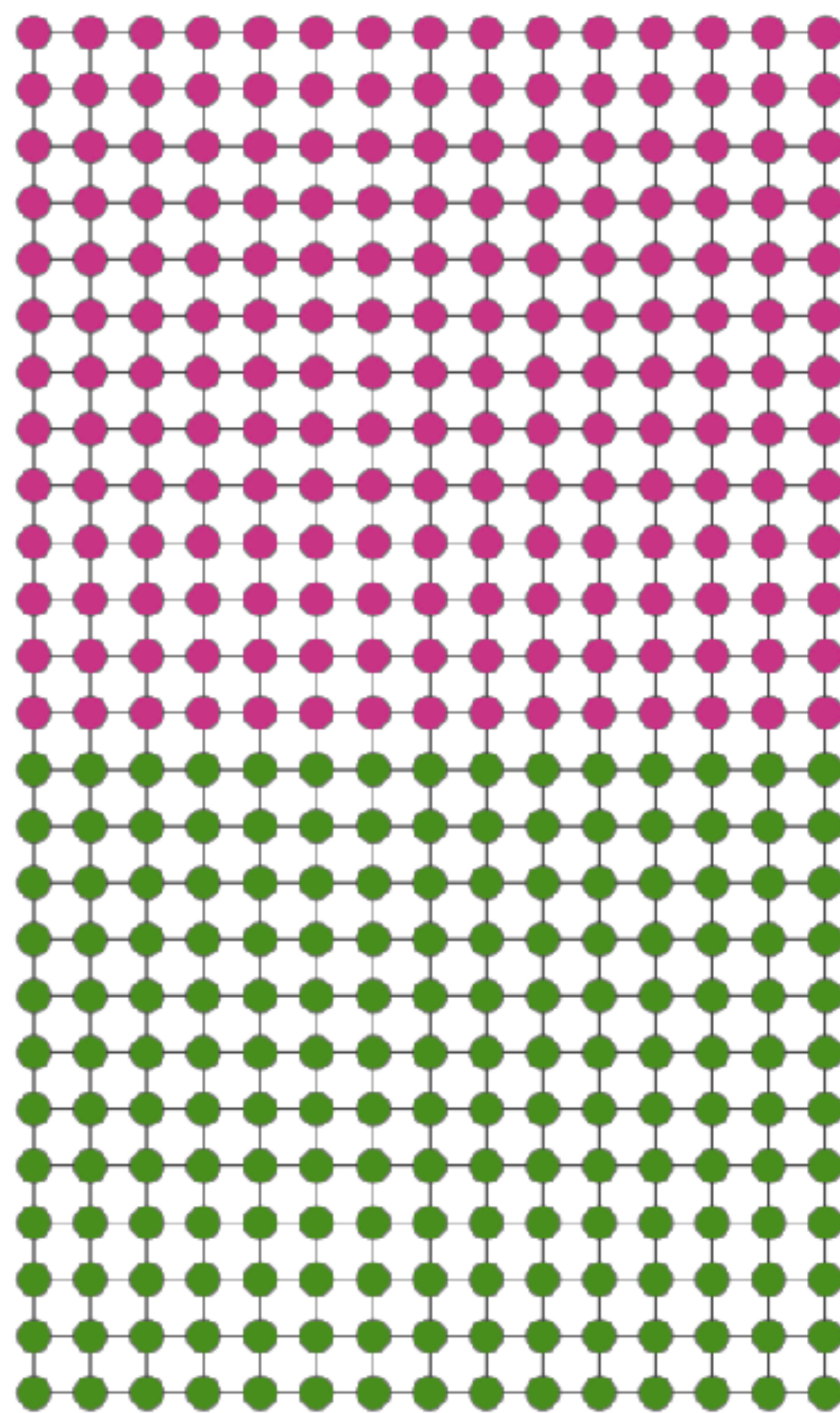
Moran's I

- It's almost the definition of "spatial autocorrelation" in the whole field of geography and spatial statistics
- It comes standard in ArcGIS
- **Positive** means segregated
- **Negative** means checkerboarded
- **Zeroish** means random



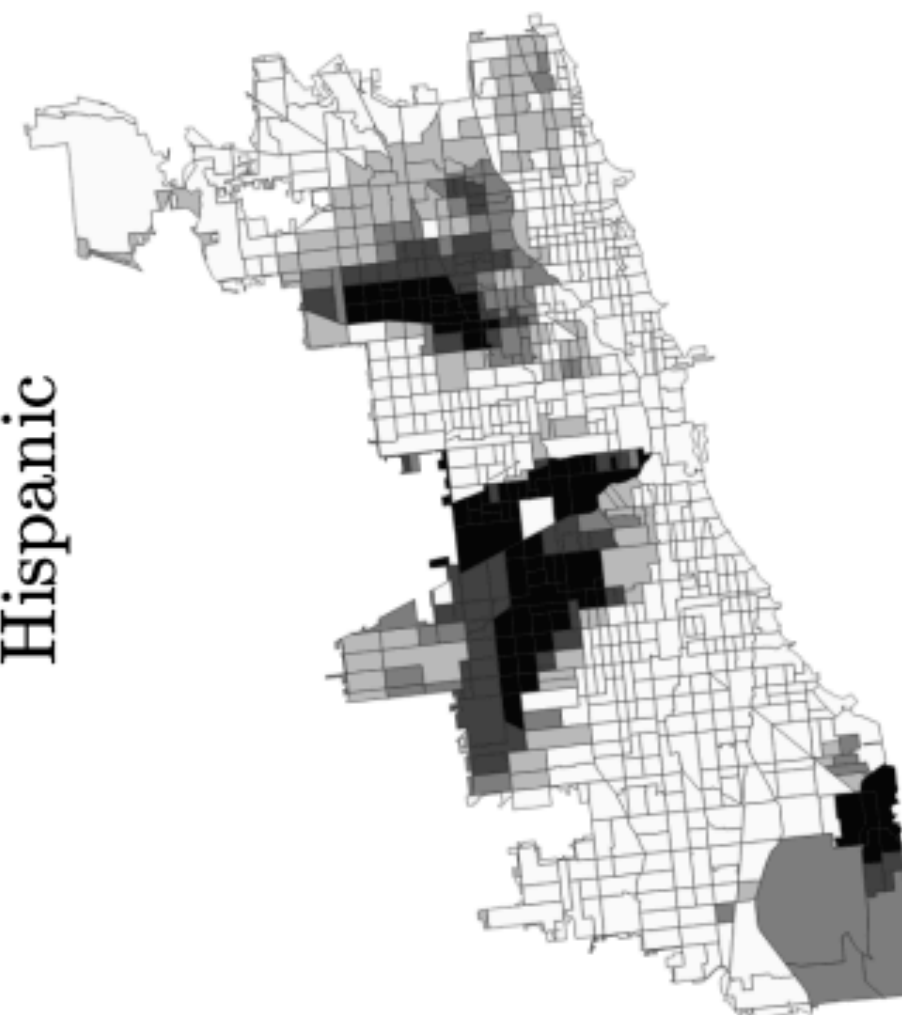
- **Moran's I:**
$$I(\mathbf{x}) = \frac{\mathbf{v}^T \mathbf{W} \mathbf{v}}{\mathbf{v}^T \mathbf{v}}$$



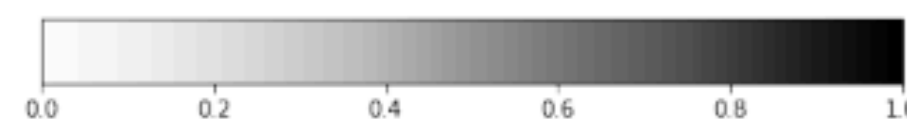
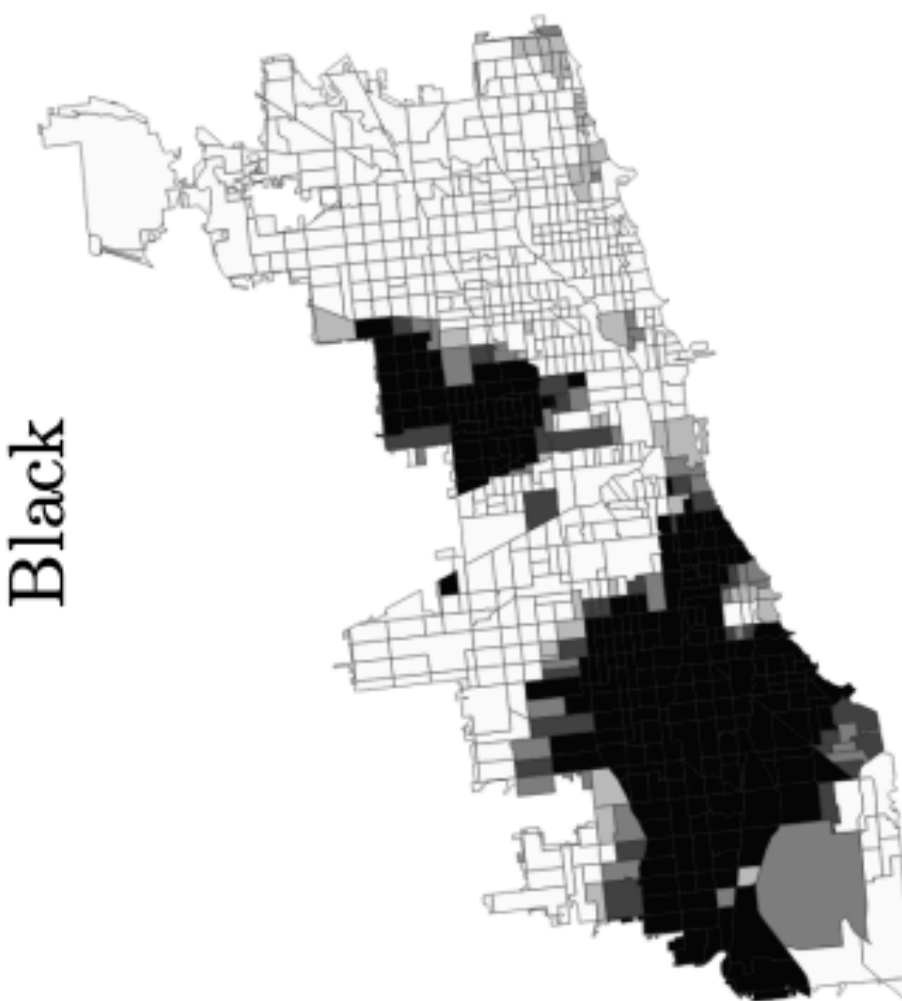


cool fact: I is the **slope** of the line that compares shares in a unit to shares in neighboring units

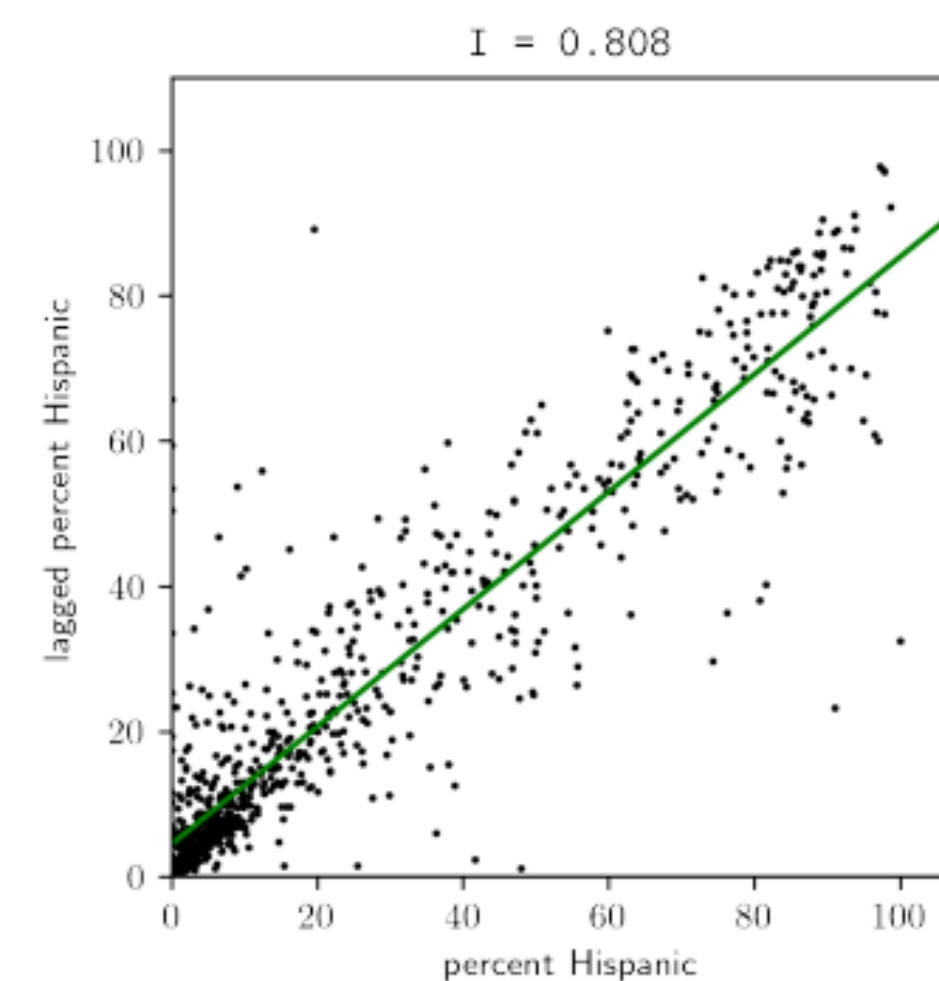
Hispanic



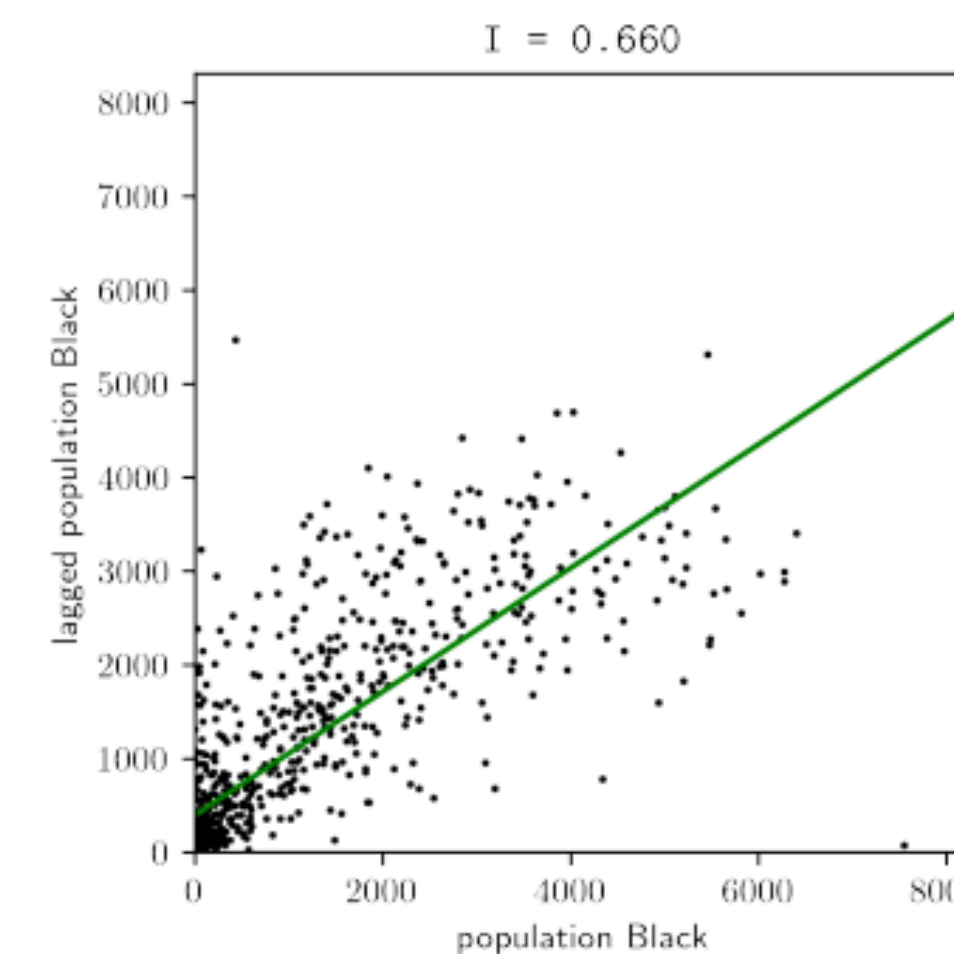
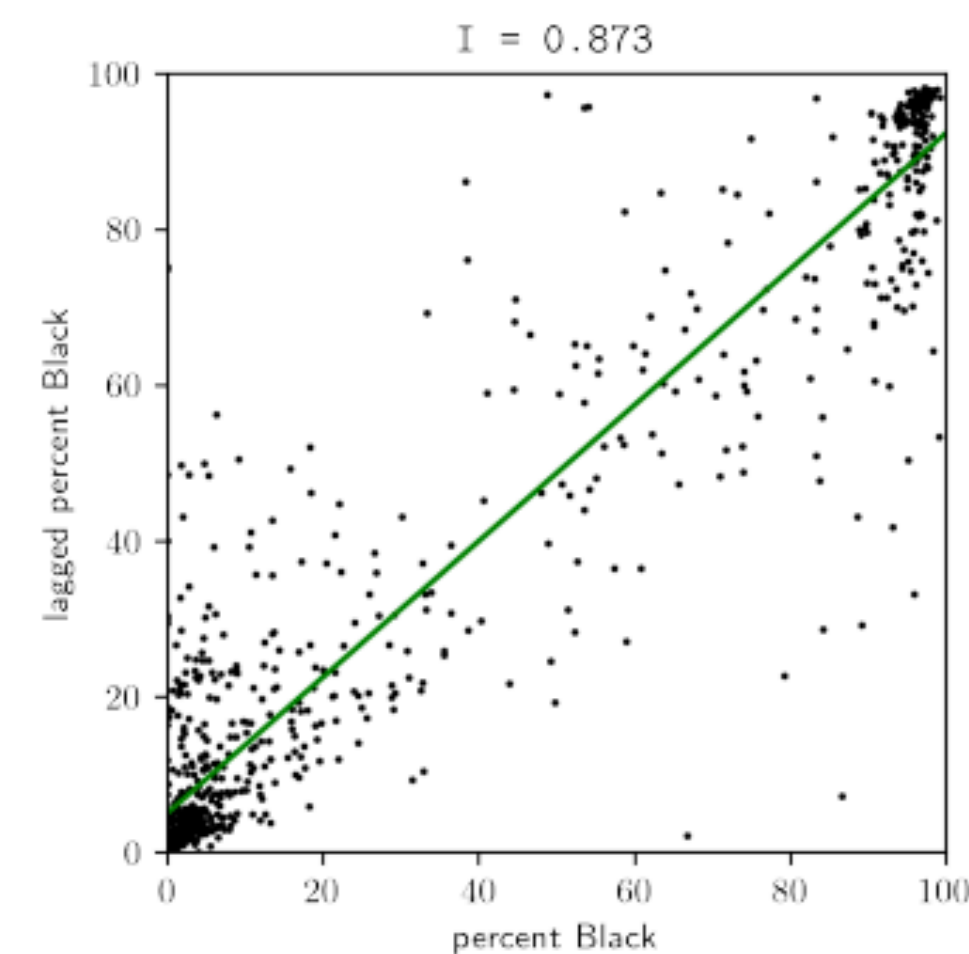
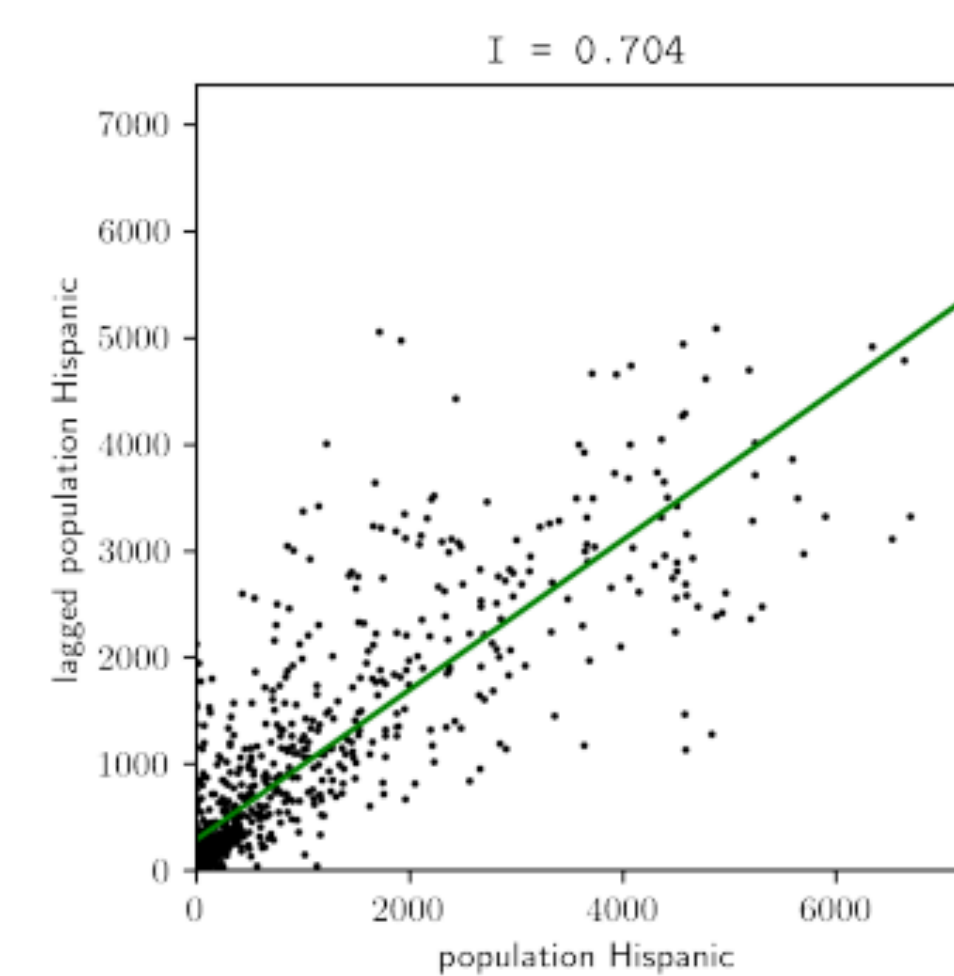
Black

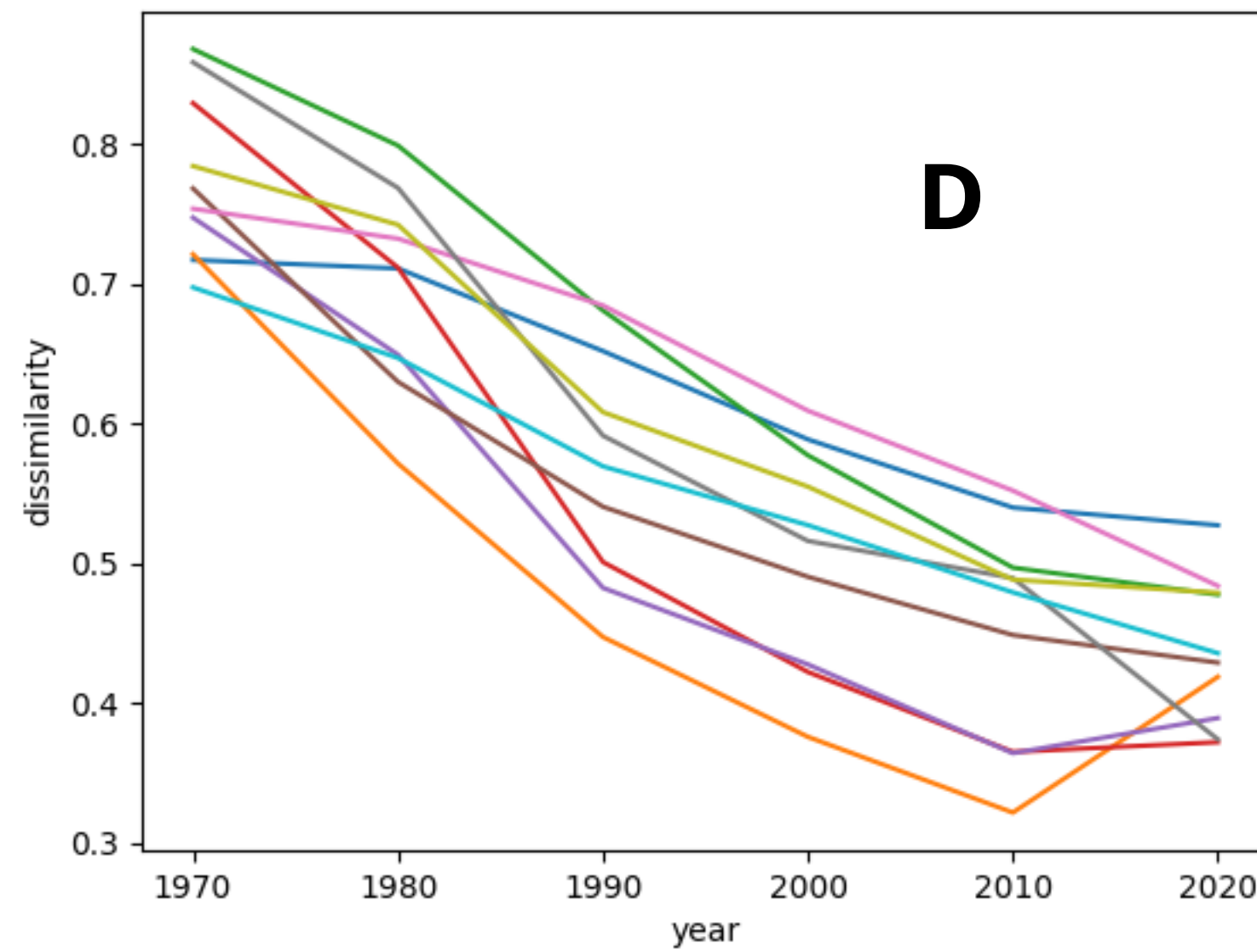


By share



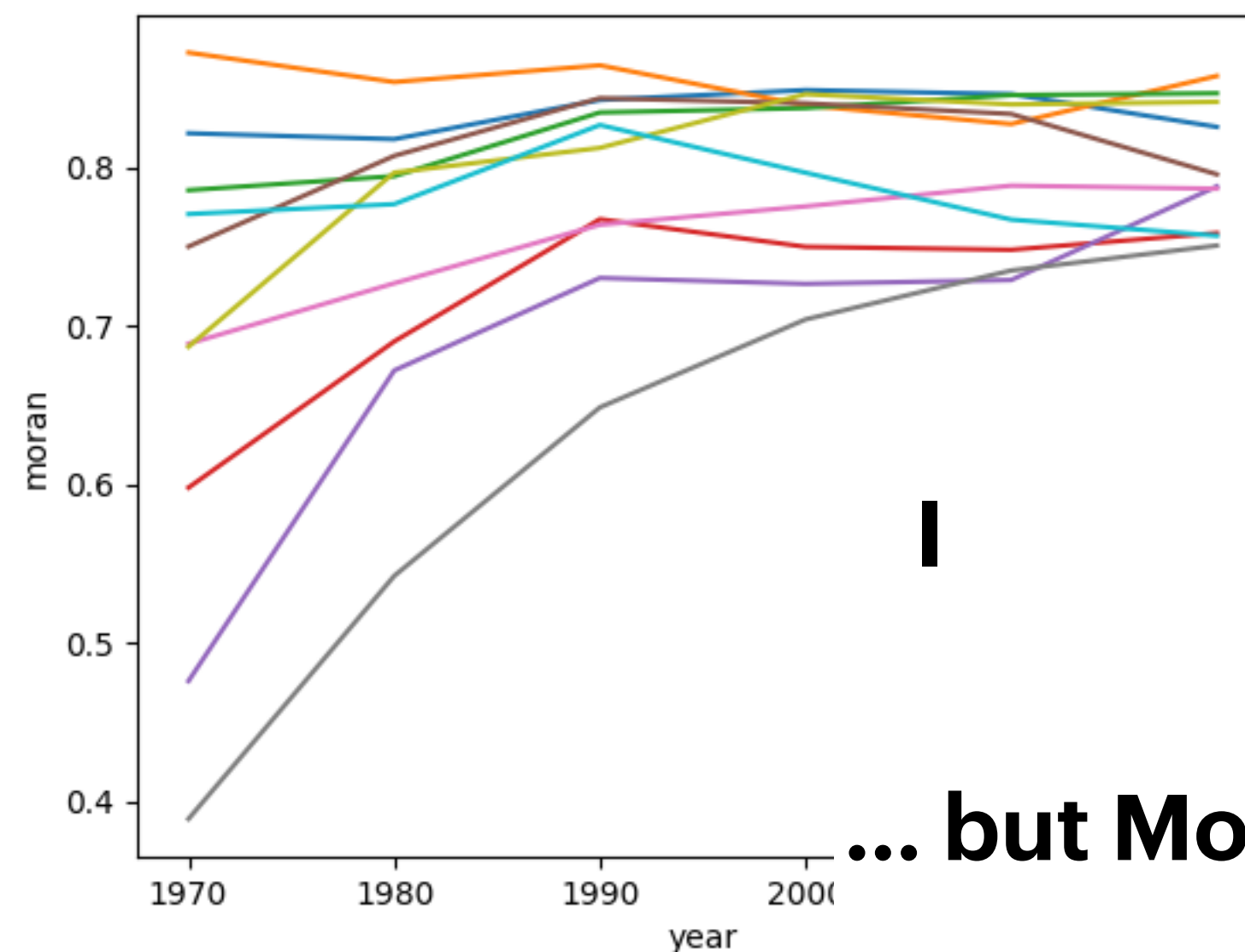
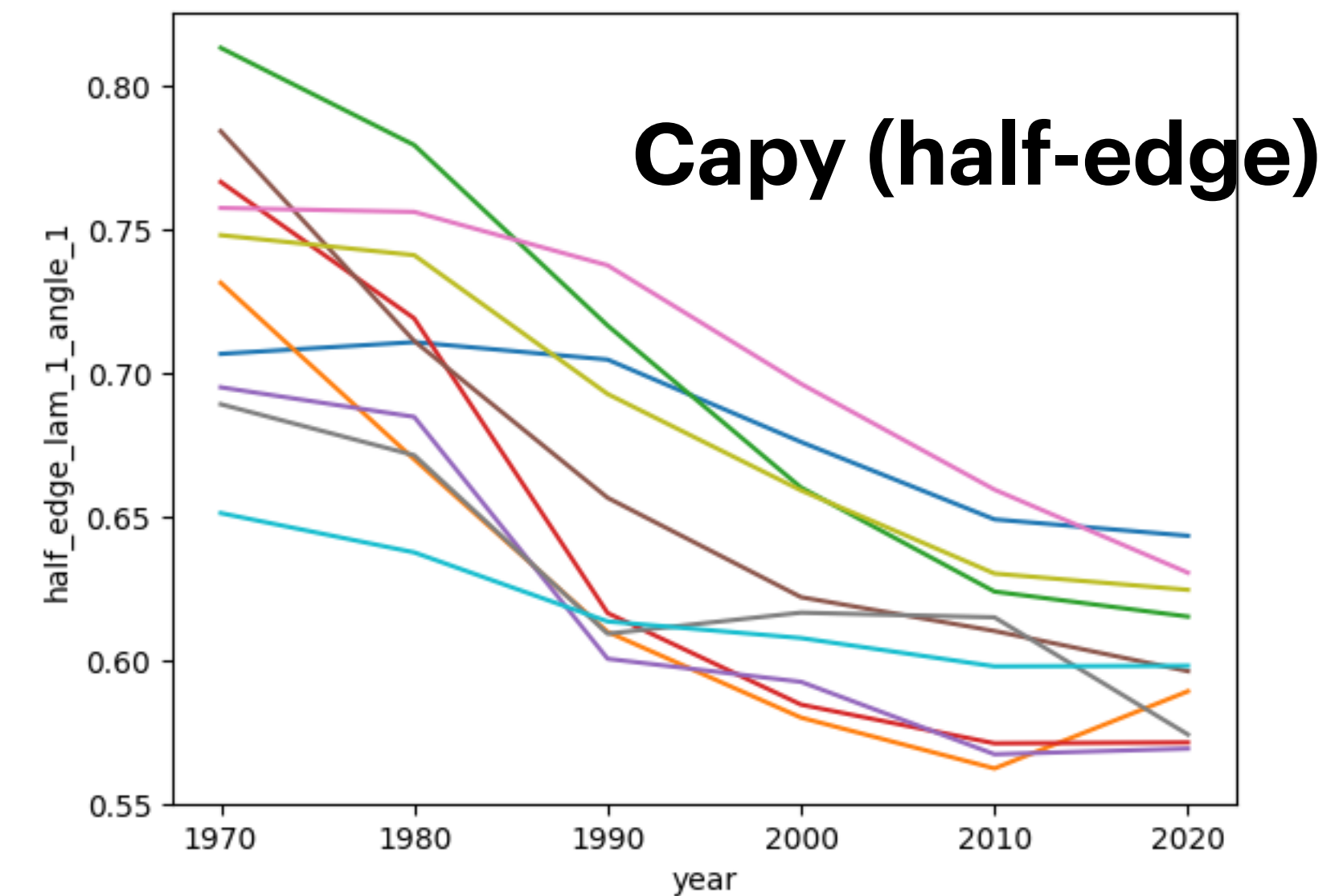
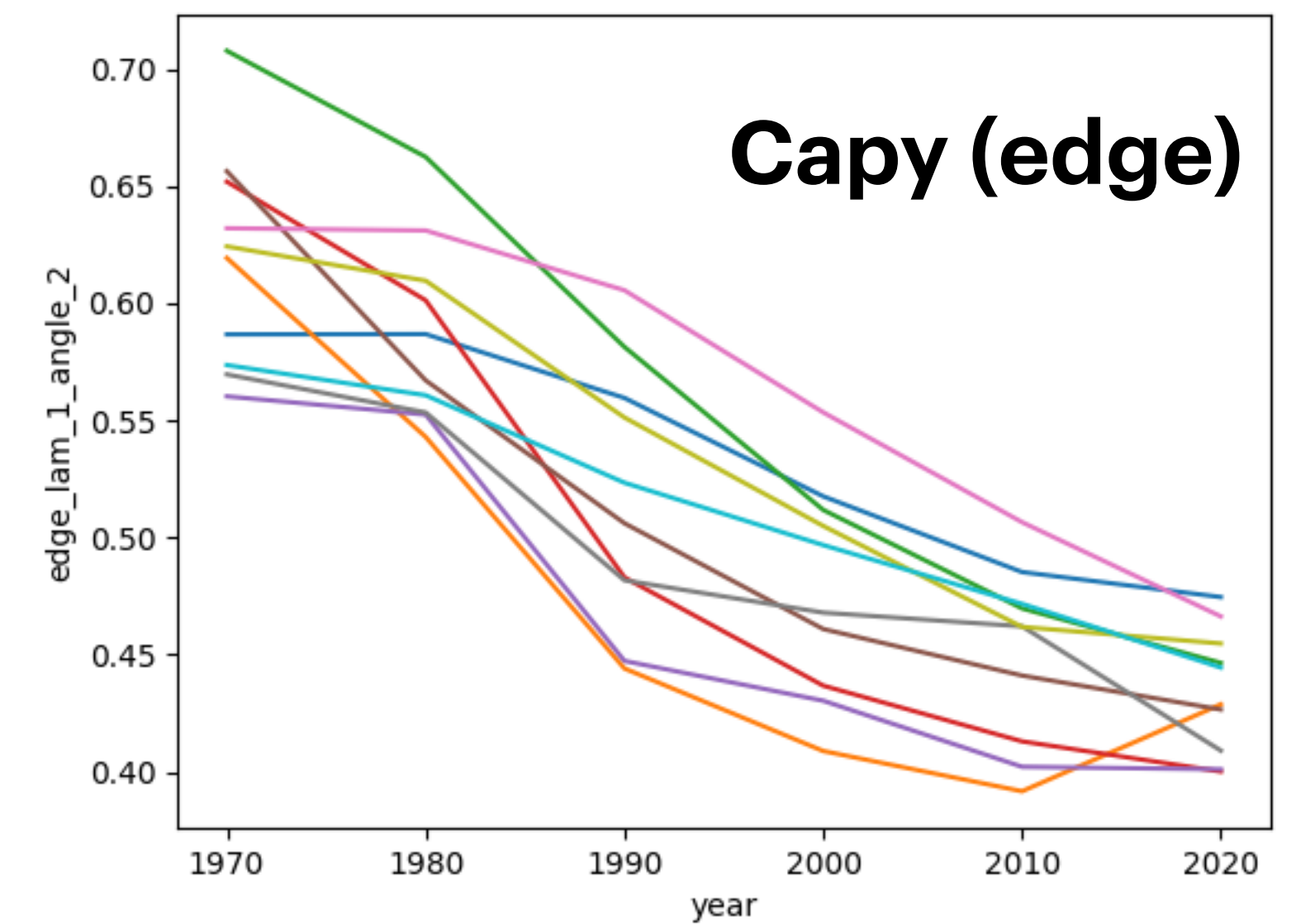
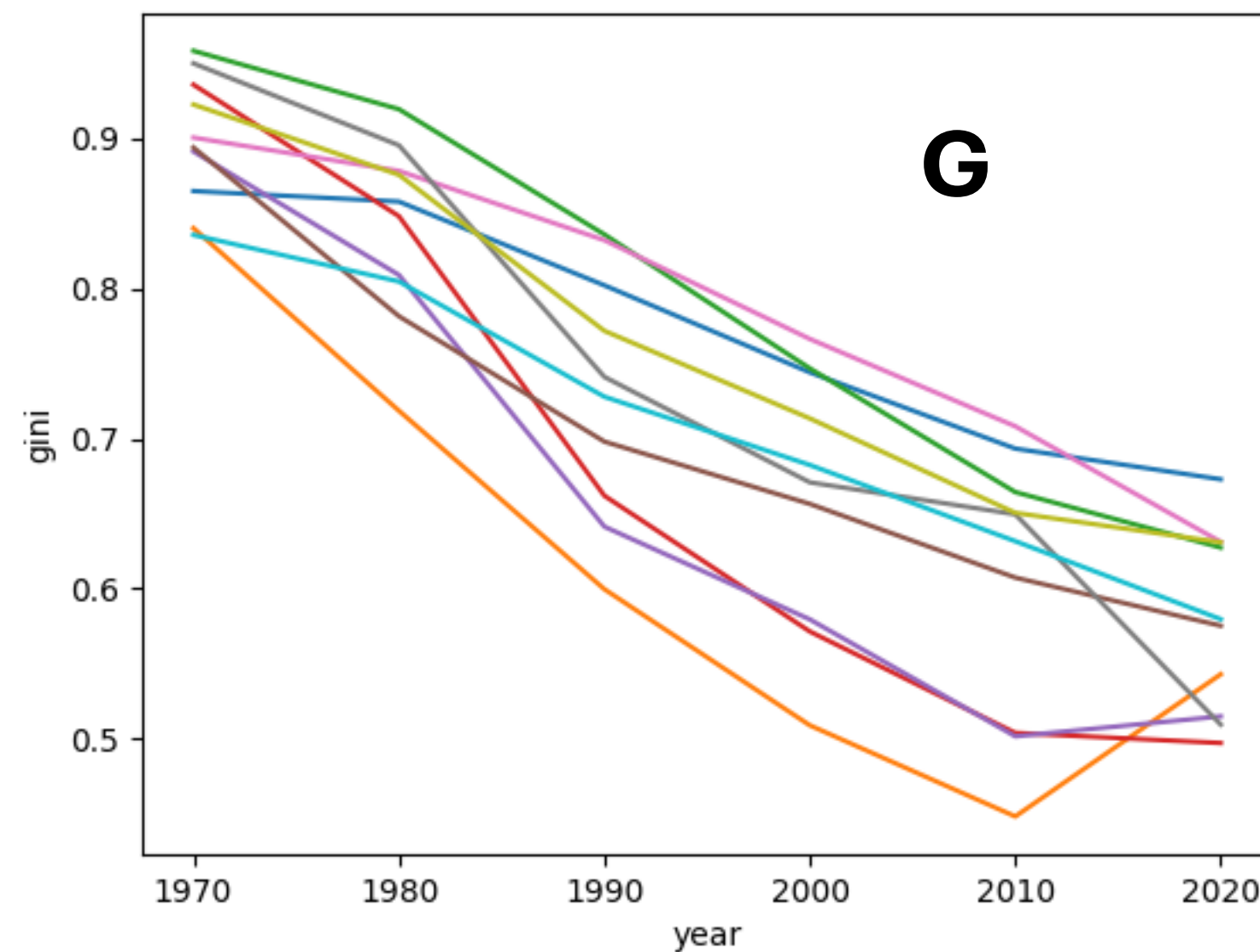
By count





There are lots of open questions on the clustering scores.

One example: almost all of the scores agree that U.S. metros are getting less segregated over time.



... but Moran's I says the opposite!

